LOT Pager (LOT-HW-01) — Product Plan

Document: LOT-HW-01 / PLAN · **Rev:** A (2026-11-13) · **Owner:** Vadik, Inventor — COSMO® CIA **Status:** Concept → EVT planning

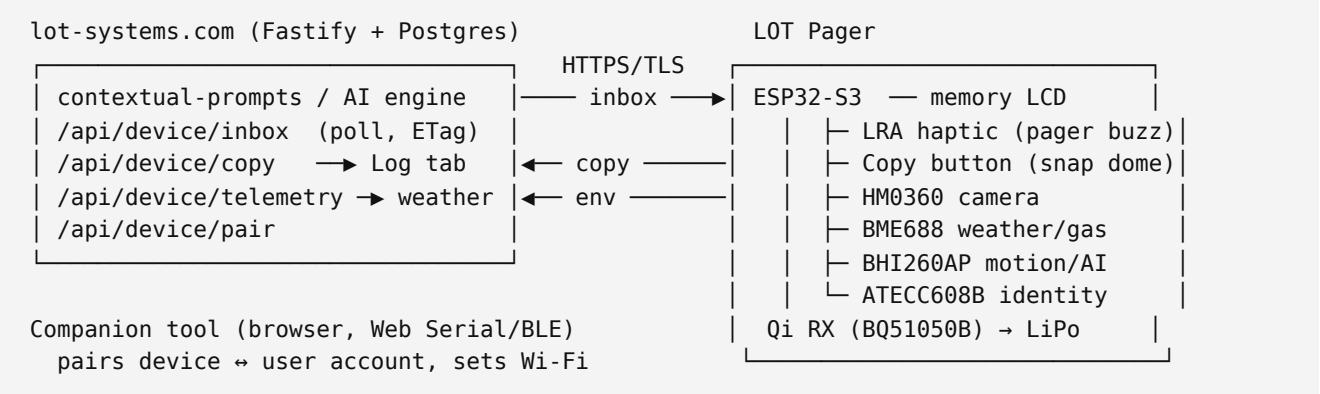
1. One-line definition

A 40 × 40 × 5 mm stainless-steel square that sits on a desk or in a pocket. It receives short, autonomous notices from lot-systems.com ("Coffee time!"), shows them on a small silver screen, and has one button — **Copy** — which sends the current notice (and optionally a photo and the local weather reading) back to the user's **Log** tab on lot-systems.com.

2. Requirements trace (from the brief)

#	Brief item	How the plan meets it	Doc
1	PCBWay	PCB fab, PCBA (100 units), CNC 316L steel and mirror polish all quoted at PCBWay as one vendor	08
2	Pager-like notification from AI site	Server pushes AI notices to a device inbox; device wakes, buzzes (LRA), shows text	05, 04
3	2-part stainless body	Part A: polished back shell. Part B: front frame with glass window. Joined by 4 × M1.2 screws or press-fit + adhesive	03
4	Flat silver square 40 × 40 × 5 mm	Target envelope. 5.0 mm is a stretch goal; EVT built at 6.0 mm (see §6)	03
5	Camera	Low-power AI-vision camera (Himax HM0360 class) behind the front glass	02
6	LOT API connector	New `/api/device/*` routes with per-device token; spec in doc 05	05
7	PDF manuals	Every doc in this folder builds to PDF via `scripts/hardware/build-manuals.sh`	pdf/
8	Compress info each session	Device side: CBOR + deflate batched payloads. Docs side: each session report ends with a compressed state block	04, sessions/
9	Firmware documents	Firmware spec	04
10	Software to connect with firmware	Companion provisioning tool (Web Serial + BLE) + server routes	06, 05
11	Separate documents	This folder: one file per discipline	00
12	Charger	Qi receiver on device + off-the-shelf Qi pad in box	02, 03
13	100-unit run	Manufacturing plan, test jig, QA, costing for 100	08
14	Weather sensor	Bosch BME688 (T / RH / pressure / gas, AI-trainable) behind a vent membrane	02
15	AI-grade off-the-shelf sensors	BME688 (BME AI-Studio), BHI260AP self-learning IMU, HM0360 AI camera	02
16	Copy button → Log tab	Button press → `POST /api/device/copy` → Log entry, event `device_copy`	05
17	One side polished steel	Part A back shell, #8 mirror polish	03
18	Other side: camera, screen, button	Front face: glass window over memory LCD + camera; steel key with snap dome	03
19	Wireless charger	Qi (BPP 5 W) receiver, TI BQ51050B; coil behind the **front glass**	02, 03

3. System architecture



Connectivity choice: Wi-Fi direct to lot-systems.com (no phone app needed — matches the "pager" idea). BLE is used only for first-time setup. Device polls the inbox on a schedule (default every 10 min, plus on button press and on motion wake). True push (persistent socket) is a later option; polling keeps battery life predictable.

4. Interaction model

Action	Result
New notice arrives	2 short buzzes, text on screen, LED-less (screen only)
Short press **Copy**	Current notice + weather snapshot copied to Log tab. Screen shows `COPIED ✓`
Long press (1.5 s)	Take photo → attached to Log entry
Double press	Show local weather (BME688)
Pick up (motion wake)	Screen shows last notice + time
Placed on charger	Screen shows battery %, syncs, then sleeps

5. Key engineering risks (summary — full list in doc 07)

1. **Steel blocks RF and Qi.** A full stainless shell is a Faraday cage and heats in a Qi field. The Wi-Fi/BLE antenna and the Qi coil must both sit behind the non-metal front glass. So the device charges **face-down** (glass side on the pad) — or the back gets a ceramic insert. Decision needed (doc 07, D-1).
2. **5 mm thickness.** Z-stack totals ~5.9 mm with a 1.5 mm battery (~80 mAh). 5.0 mm needs a custom 1.0 mm cell and a ≤ 2.2 mm camera. EVT at 6.0 mm, DVT tries 5.0.
3. **Radio certification.** Selling 100 units needs FCC/CE. A pre-certified module saves US$10–20 k of testing but is thicker. See doc 07, D-3.
4. **Weather sensor in a sealed steel box** reads the device, not the room. Needs a vent with ePTFE membrane and firmware self-heat compensation.

6. Feasibility verdict

Buildable with off-the-shelf parts. The look (silver square, one button, one screen) is achievable. The **5 mm** spec and **wireless charging through steel** are the two items that need a design decision before PCB layout starts. Neither blocks EVT.

7. Estimated cost (100 units)

Bucket	Per unit	×100
Electronics BOM	~$72	~$7,200

Bucket	Per unit	×100
PCB + PCBA (PCBWay, 4-layer HDI 0.6 mm)	~$18	~1, 800+ 350 setup
Stainless enclosure, 2 parts, CNC + mirror polish	~$38	~$3,800
Cover glass (printed, AR)	~$6	~$600
Qi pad + USB-C cable + box + printed manual	~$14	~$1,400
Unit total	**~$148**	**~$14,800**
NRE: EVT/DVT prototypes, stencils, test jig, CNC fixtures		~$4,000–6,000
Certification (FCC + CE/RED), if not modular		~$12,000–20,000

All prices are planning estimates (Nov 2026), not quotes. Doc 02 has per-part numbers and links.